

Digital Product Passports (DPP) and Sustainability Solutions

Kim Eun-hie (Oracle Korea Consultant)

The barriers to entering the European market are rising. Beginning in 2027, the European Union will fully implement the **Digital Product Passport (DPP)** framework. Products that fail to comply may face restrictions not only at customs but also in the European marketplace itself. Under the DPP, information such as a product's country of origin, manufacturer certifications, hazardous substance content, durability, recyclability, and carbon emissions will be transparently disclosed to consumers through QR codes and APIs.

The real challenge, however, lies in integrating fragmented data across existing ERP, PLM, and SRM systems into a DPP-compliant framework while building the IT infrastructure needed for every participant in the supply chain to securely share certified information.

The first key to implementing DPP is product identification and information accessibility. Consumers must be able to access complete product information immediately by scanning a QR code. This is where circular fashion cloud services such as **Care ID**, developed by Yoonhoe Co., Ltd., have gained attention. Care ID assigns a unique digital identity to every garment. Once a brand enters information such as fiber composition, manufacturing facility, and production date, the platform verifies the data and generates a globally standardized QR code.

After scanning the code, consumers can authenticate their purchase and register themselves as the product's owner through a digital signature. This enables proof of authenticity and global traceability throughout the product's lifecycle, including future resale transactions.

The second pillar is AI-powered supply chain transparency. **Retraced**, a German supply chain transparency platform, helps fashion companies visualize supply chains extending beyond second-tier suppliers, an area that approximately 95% of brands lack visibility into. The platform enables companies to detect labor exploitation and environmental violations at the fabric, dyeing, and raw material stages of production.

Supported by Oracle, Retraced was adopted by **Shinwon** in September 2024 and is now used by more than 150 brands, including **Desigual** and **Victoria's Secret**. It also helps companies prepare for the European Union's **Corporate Sustainability Due Diligence Directive (CSDDD)** by verifying that products are free from ethical compliance risks throughout the supply chain.

Retraced's supply chain mapping process is straightforward. Brands invite their first-tier suppliers onto the platform, who in turn invite their own suppliers, extending visibility through third- and fourth-tier raw material providers. The platform visualizes the entire supply chain on an interactive map, while each supplier uploads certifications and supporting documentation.

These records are stored in blockchain tables within Oracle's Autonomous Database, helping protect data integrity against tampering or alteration.

The platform's AI capabilities further strengthen supply chain governance. Instead of relying on SQL queries, users can search data using natural language. Supplier records and certification documents are

vectorized and indexed, allowing AI to quickly retrieve relevant information while identifying duplicate or expired records.

AI also automatically extracts information from supplier-submitted PDF documents, verifies certification validity against external databases, and generates regulatory compliance reports. In addition, geospatial analytics continuously monitor whether raw material sources are located near areas associated with deforestation or regions considered at high risk for forced labor. Rather than simply mapping the supply chain, these capabilities enable meaningful, proactive risk management.

Ultimately, the fashion industry should view Digital Product Passports not merely as another regulatory requirement, but as an opportunity for digital transformation. Companies that adopt proven technology solutions while implementing API-driven integration quickly and strategically will be best positioned to compete in the DPP era beginning in 2027.